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Ivar Cashin Eriksson

Dear Dr Tsaousoglou and Members of the Selection Committee,

I am applying for the PhD scholarship in AI Agents in Electricity Markets at DTU Compute. After seven years working with machine learning in research and industry, I want to return to research to spend more time developing methods and investigating the tough questions behind them. This project combines my interests in machine learning, mathematics, and economics. Having worked with automated trading and price forecasting in a highly regulated pharmaceutical environment, I am particularly interested in how individually optimised trading strategies interact, which behaviours emerge, and how regulation can influence those outcomes.

Alongside my studies, I worked as a Trading Supervisor at Ampfield in Stockholm, supervising automated trading systems and managing trading risk. I also contributed implementations of optimisation algorithms for the automated trading systems. This gave me practical exposure to automated decision-making in financial markets.

After graduation, I spent time researching methods for automated radiotherapy planning as a full-time researcher at RaySearch Laboratories. I used autoencoders to represent patient anatomy and sparse Gaussian processes to predict radiation dose distributions. I also worked on optimisation methods to better reflect clinical priorities. Working with medical physicists and clinicians taught me to examine how modelling choices relate to the decisions a system needs to support. I enjoyed the combination of mathematical work, practical constraints, and discussion with specialists.

Since then, I have taken responsibility for larger ML systems. At Valcon, I led projects from problem formulation through deployment, including a real-time neural-network pipeline combining text and categorical data. At Signum Life Science, I now lead the architecture and development of a forecasting platform supporting thousands of independently trained models. This involves data validation, reproducible evaluation, experiment tracking, and batch inference. These roles have taught me to structure complex software, investigate failures, and make technical decisions together with colleagues and domain experts.

My MSc in Engineering Physics, specialising in Statistical Learning and Data Analytics, and my additional BSc in Economics provide complementary foundations for this project. My studies included Reinforcement Learning and Optimisation at KTH, and Game Theory and Control at ETH Zürich. I would bring experience developing and evaluating mathematical methods, implementing them in Python, and building reproducible computational workflows. I would like to build on this background through research combining theoretical analysis with numerical experiments, while developing my understanding of electricity markets.

I welcome the opportunity to discuss the project and how my background could contribute to the research.

Kind regards,
Ivar Cashin Eriksson